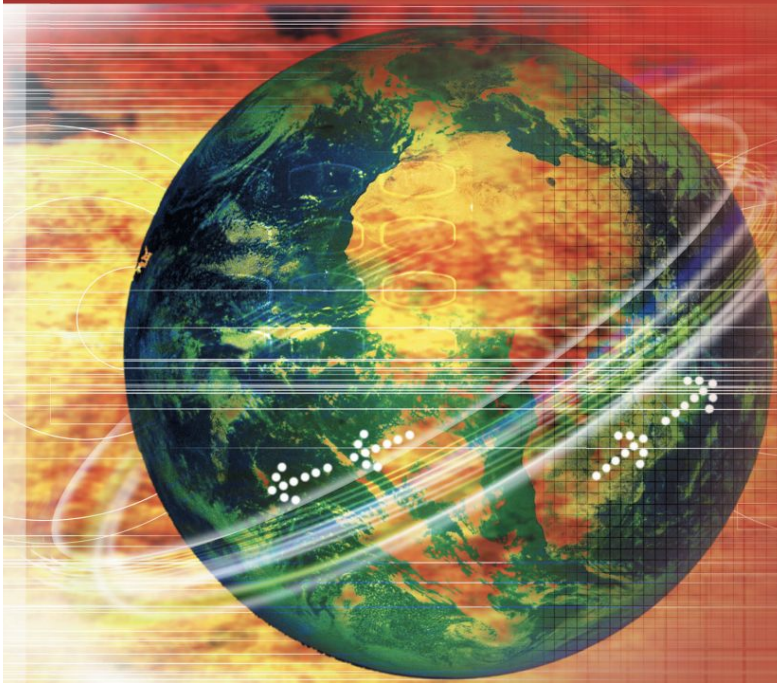


FIFTH EDITION

Data Communications AND Networking



BEHROUZ A. FOROUZAN

Chapter 9

Introduction to Data-Link Layer



The Role of the Data-Link Layer

No **define protocols** for the data-link or physical layers.

- ❑ These layers are managed by individual network technologies (wired/wireless).
- ❑ Provide foundational services to upper layers.

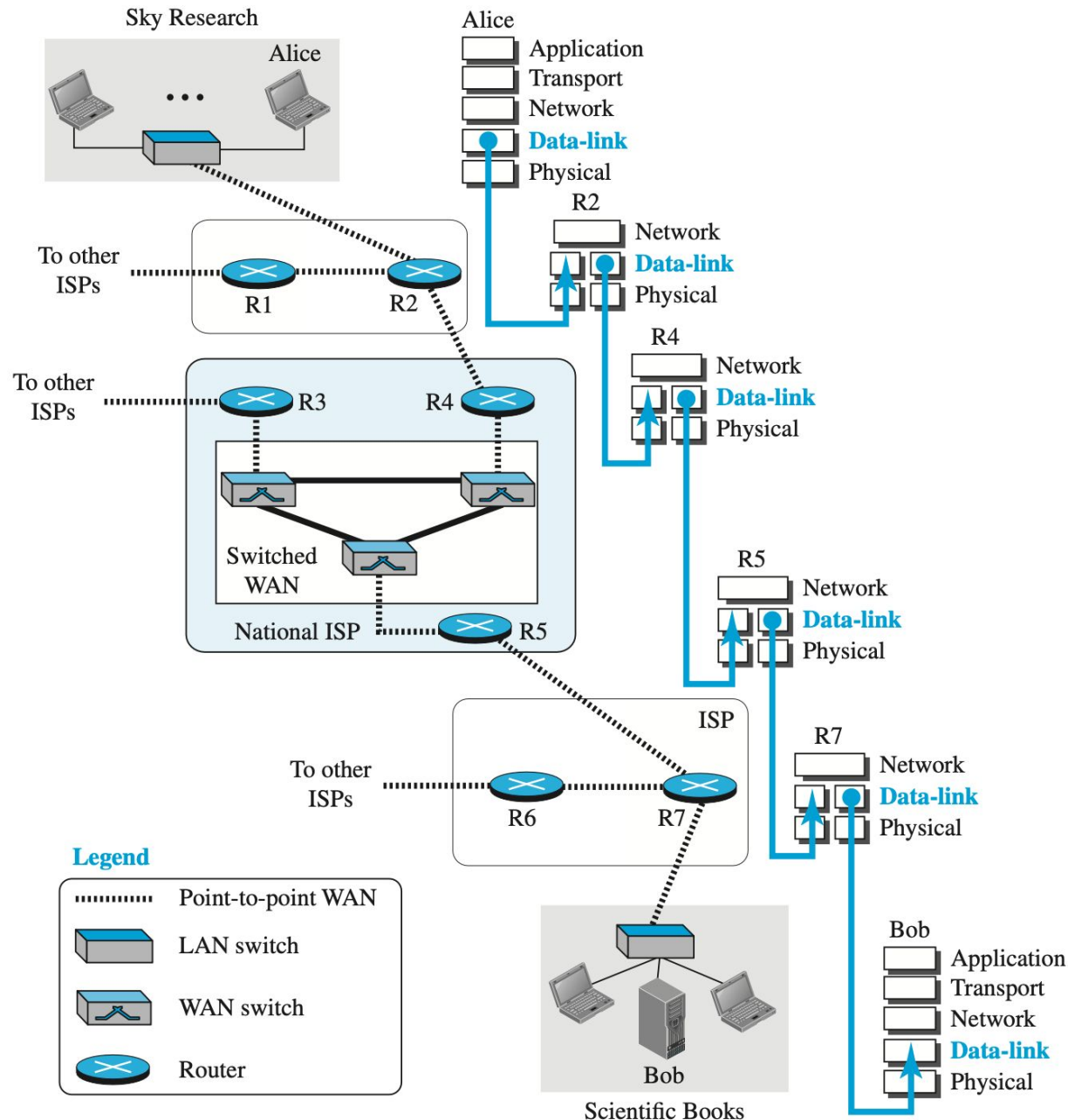
Section 1: Overview of the Data-Link Layer

- ❑ **Links and nodes**
- ❑ **Services provided** by the data-link layer
- ❑ Two **sublayers**: MAC and LLC

Section 2: Link-Layer Addressing

- ❑ The necessity for addressing at the data-link layer
- ❑ Three types of link-layer addresses
- ❑ Address Resolution Protocol (ARP)
 - ❑ Mapping of network layer addressing and link layer addressing
 - ❑ Locating next hop link layer address

Figure 9.1: Communication at the data-link layer



- ❑ The Internet consists of **multiple networks** connected by **routers or switches**
- ❑ For each network, a data link connection is required.
- ❑ If a device is connected to multiple networks, then that device is involved with multiple data link layer.
- ❑ Only one data-link layer is involved in Alice and Bob
- ❑ Two data-link layers are involved at each router. one for incoming, one for outgoing.

Nodes and Links

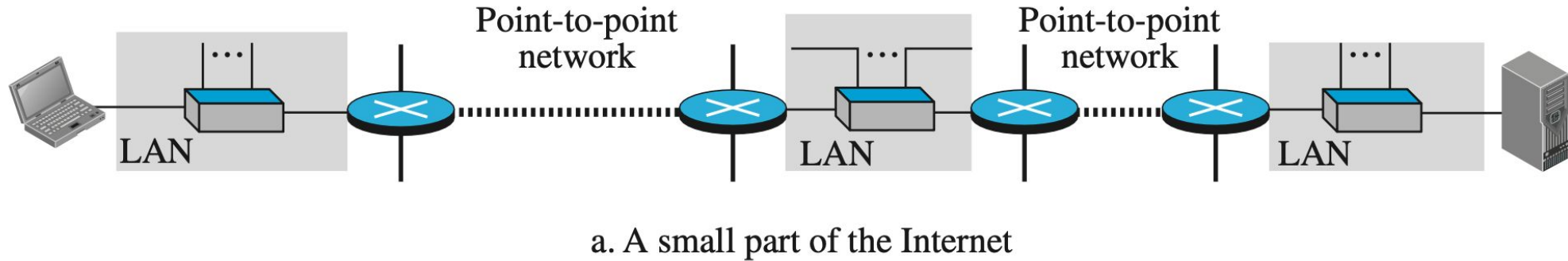


Figure 9.2: Nodes and Links

- ❑ Communication at the data-link layer is node-to-node.
- ❑ Two end hosts (Source and Destination) and the routers are *nodes* and the networks in between as *links*.

Roles of Data Link Layer

- ❑ In between Physical and Network layers.
- ❑ Provides services to network layer, uses services of physical layer.

Scope of Data Link Layer

- ❑ Node-to-Node Delivery
- ❑ Responsible for sending a **datagram to the next node** in the path
- ❑ Uses **encapsulation** (sender) and **decapsulation** (receiver)

Encapsulation and Decapsulation

Source Host:

Data-link layer encapsulates datagram into a **frame**

Frame sent to the **router**

Router:

Decapsulates the received frame

Encapsulates the datagram into a new frame

Sends the new frame to **destination host**

Note: Router typically has **3 data-link layers** (one per physical link)

Destination Host:

Data-link layer **decapsulates** the frame to extract the datagram

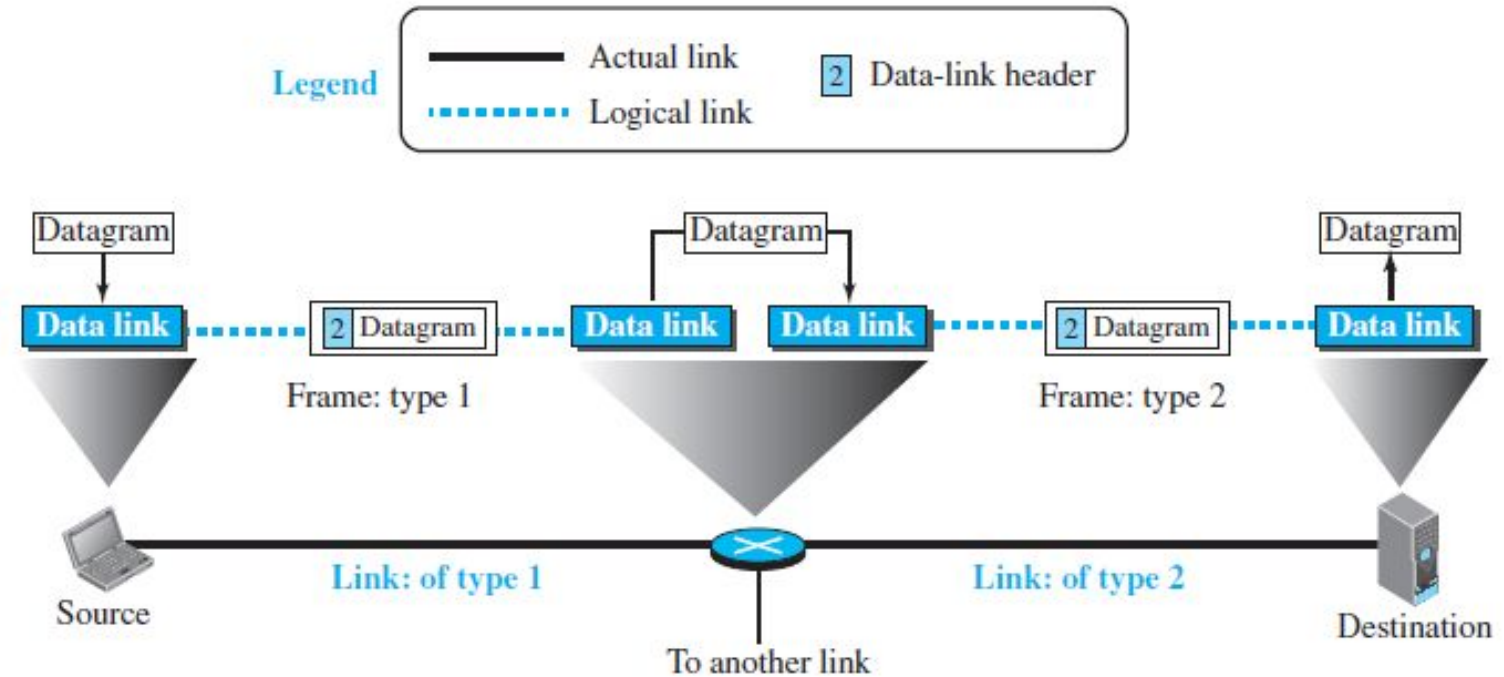


Figure 9.3: A communication with only three nodes

Services of Data Link Layer

❑ Framing

- ❑ Encapsulates network-layer datagram into a frame
- ❑ Adds header and trailer
- ❑ Format varies by protocol
- ❑ Ensures proper data boundaries

A packet at the data-link layer is normally called a *frame*.

❑ Flow Control

- ❑ Prevents sender from overwhelming the receiver
- ❑ If buffer overflows:
 - ❑ Drop frames or
 - ❑ Send feedback to pause/slow down sender
- ❑ *Used also at transport layer*

Services of Data Link Layer (2)

☐ **Error Control**

- ☐ Handles errors from physical transmission
- ☐ Steps:
 - ☐ Detect errors
 - ☐ Correct at receiver or
 - ☐ Request retransmission
- ☐ Critical for reliable communication

☐ **Congestion Control**

- ☐ Congestion in link results in frame loss
- ☐ **Rarely implemented in data-link layer**
 - ☐ Handled mostly in network & transport layers

Two Sublayers

- ❑ **Data Link Control (DLC) Sublayer**
 - ❑ Handles functions **common to all link types** (e.g., framing, flow control, error control)
 - ❑ Used in both **point-to-point** and **broadcast** links
- ❑ **Media Access Control (MAC) Sublayer**
 - ❑ Deals with issues **specific to broadcast links** (e.g., controlling access to a shared medium)
 - ❑ Crucial in **LANs** and wireless networks

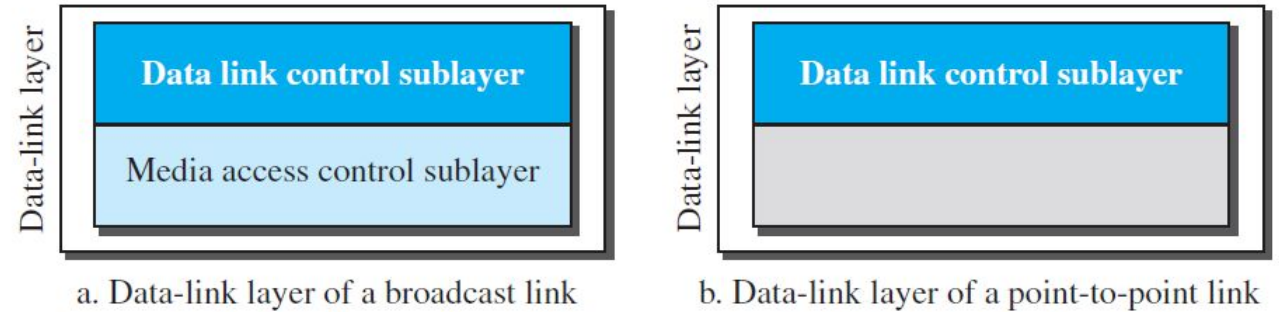


Figure 9. 4: Dividing the data-link layer into two sublayers

Link Layer Addressing

- ❑ IP addresses identify source and destination hosts in the Internet
- ❑ Internet is connectionless. Datagrams may take different paths.
- ❑ IP addresses must not change during transmission

- ❑ Thus Link Layer addressing is used deliver frames over each individual link.

Link Layer Addressing

- Link Layer Address is also known as MAC address, Physical address.
- Unique **per node on a link**.

Every time a datagram is passed to a new link, **new link-layer addresses** are assigned in the frame, even though the **IP addresses remain unchanged**.

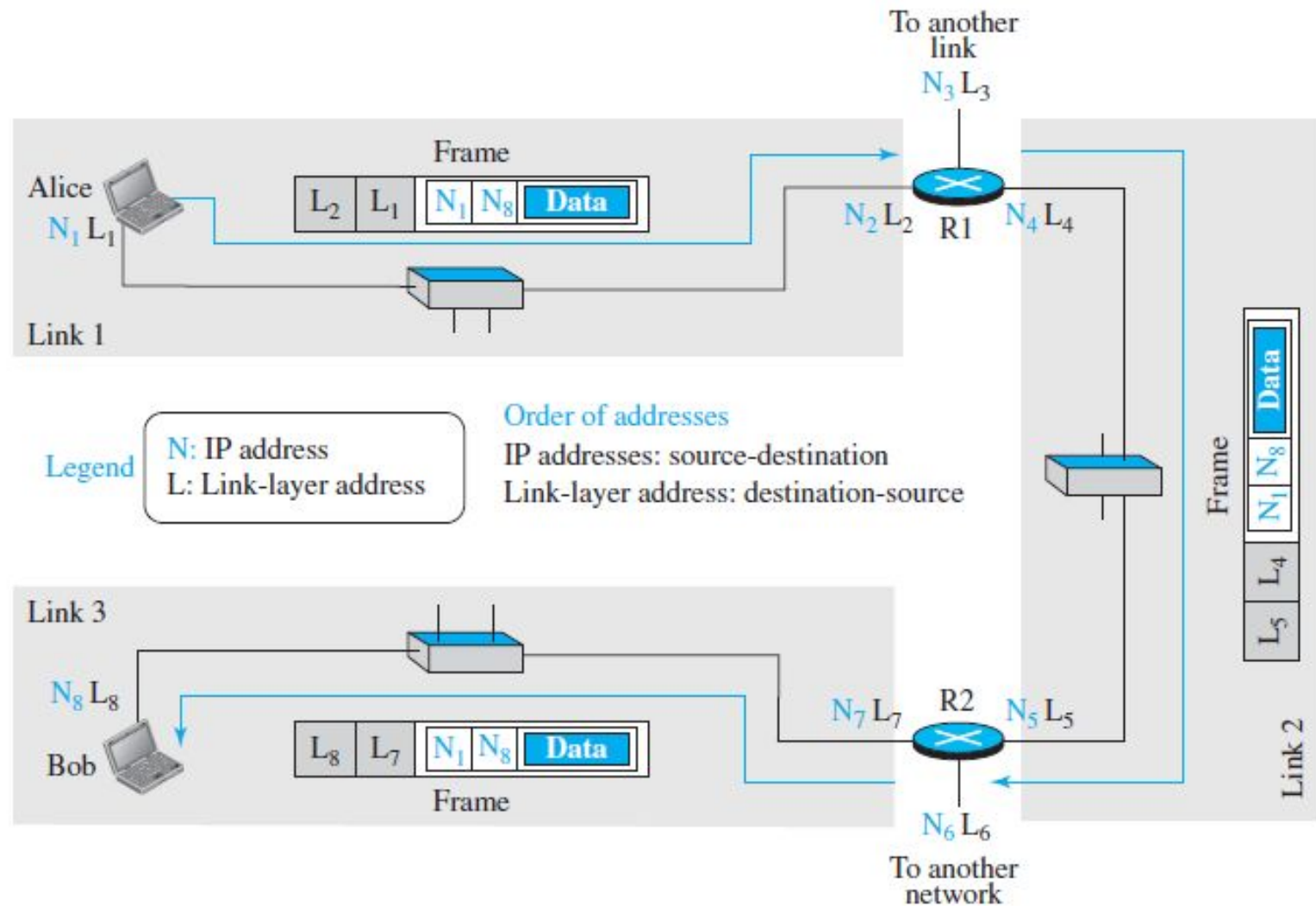


Figure 9. 5: IP addresses and link-layer addresses in a small internet

Link Layer Addressing

- ❑ Q1: Why do routers need IP addresses if they don't appear in datagrams?
 - ❑ Routers do appear as senders/receivers in routing protocols (e.g., RIP, OSPF)
- ❑ Q2: Why does each router interface need a separate IP address?
 - ❑ Each interface is connected to a different link
- ❑ Q3: How are source & destination IP addresses set?
 - ❑ Source host uses Own IP as source, Destination IP found via DNS (Domain Name System)
- ❑ Q4: How are link-layer addresses determined per link?
 - ❑ Each node knows its own MAC address
 - ❑ Destination MAC address is obtained via ARP (Address Resolution Protocol)
- ❑ Q5: What is the size of link-layer addresses?
 - ❑ Depends on the data-link protocol used (e.g. Ethernet: 48 bits and Other links may use different sizes)

Types of Link-Layer Addresses

☐ Unicast Address

- ☐ Assigned to a **single host or router interface**
- ☐ Enables **one-to-one** communication
- ☐ Frame is delivered to **one specific receiver**

00-1d-a1-02-b3-c4

Unicast Address

☐ Multicast Address

- ☐ Represents a **group of devices** on the same link
- ☐ Enables **one-to-many** communication
- ☐ Frame is delivered to **all members of the group**
- ☐ Scope is **local to the link**

01-00-5e-00-00-16

Multicast Address

☐ Broadcast Address

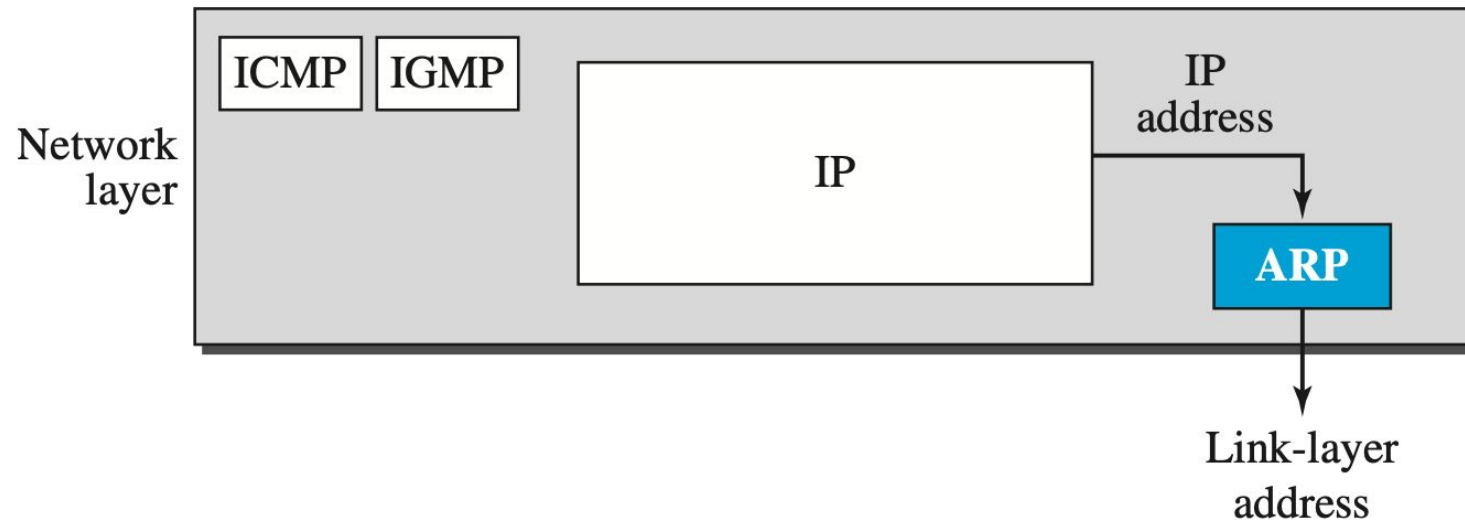
- ☐ Special address used to reach **all devices** on the link
- ☐ Enables **one-to-all** communication
- ☐ Frame is received by **every node** in the network segment

ff-ff-ff-ff-ff-ff

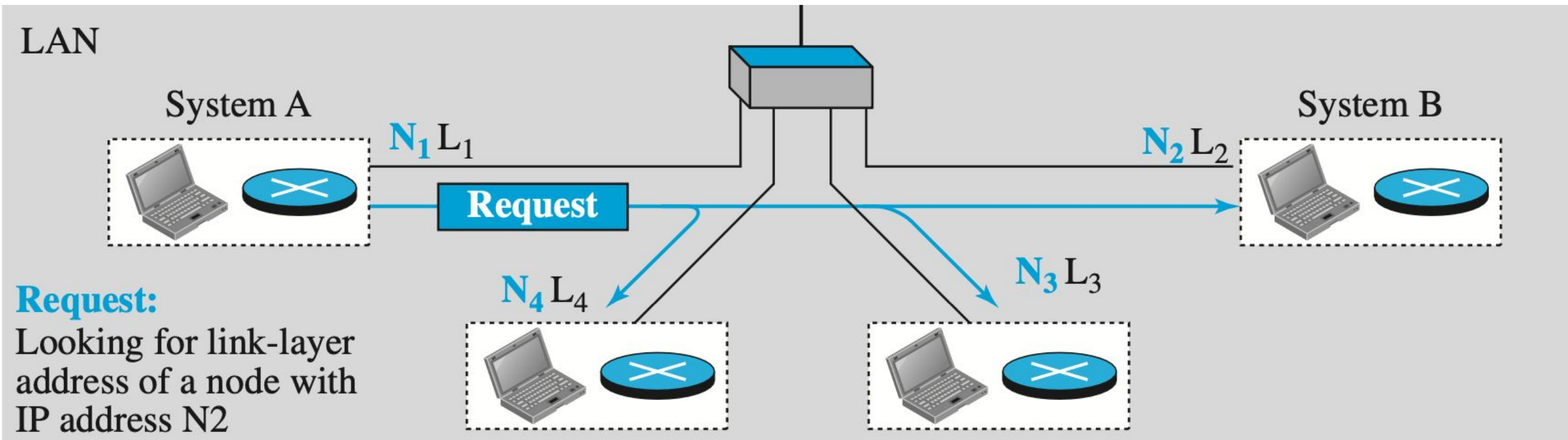
Broadcast Address

Address Resolution Protocol (ARP)

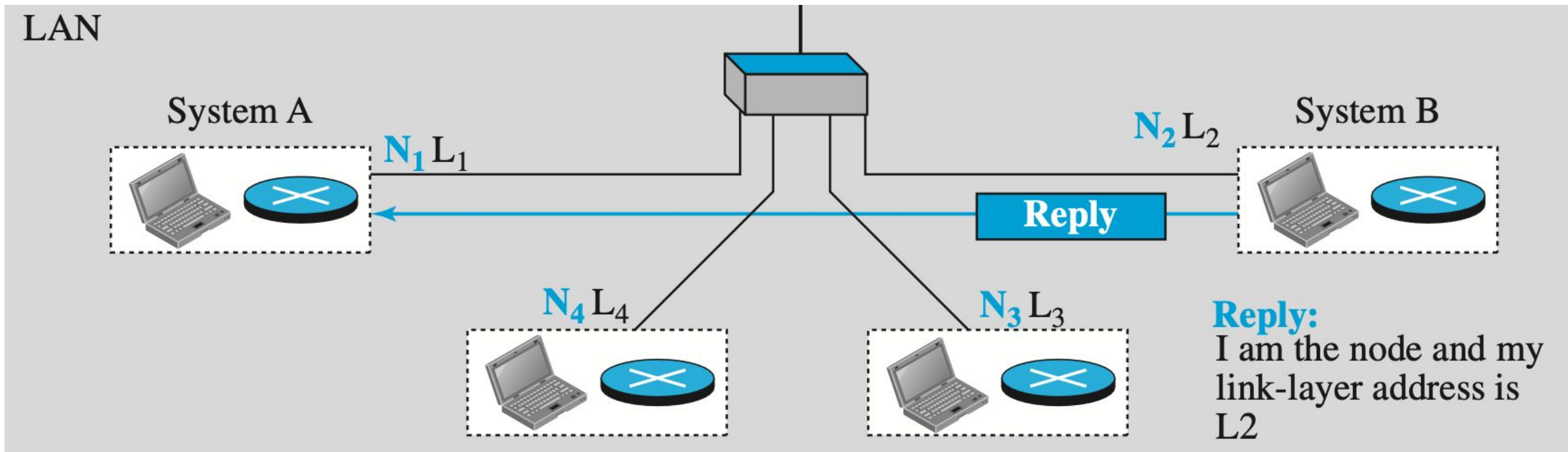
- ❑ ARP table has a mapping between IP address vs. Link Layer Address.
- ❑ This table is populated by ARP protocol.
- ❑ ARP is an auxiliary protocol. Its Layer 3 protocol.



ARP Request



ARP Reply



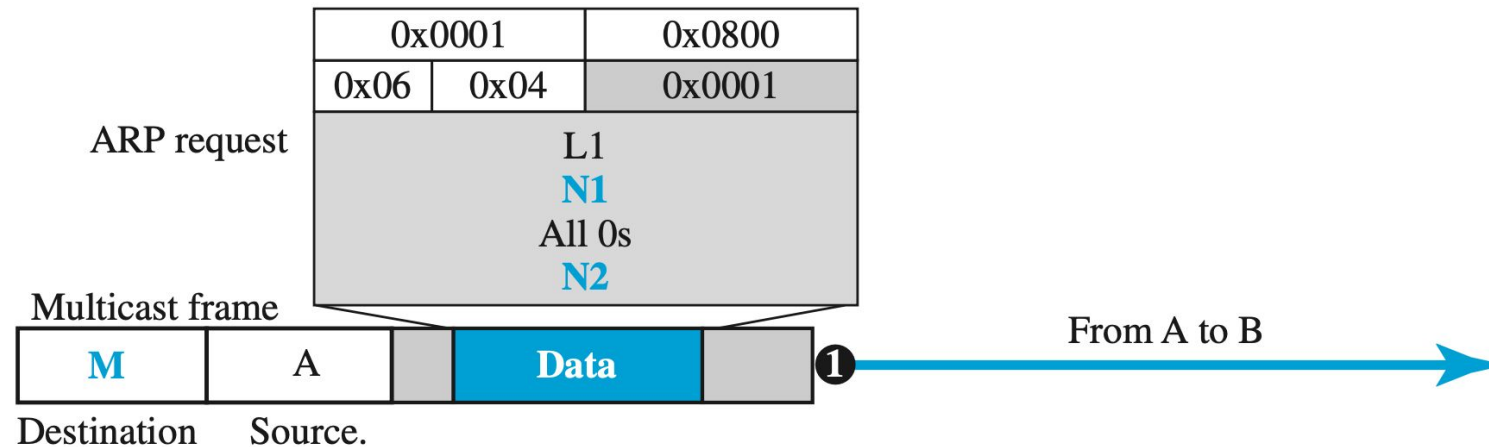
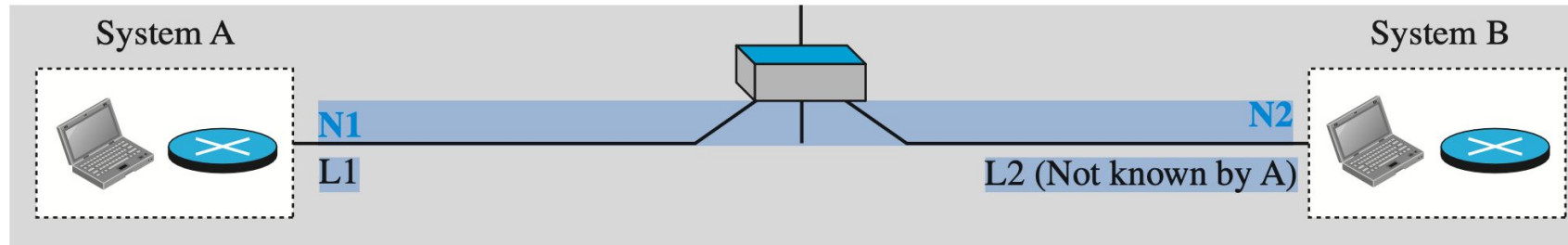
ARP Packet Format

0		8		16		31	
Hardware Type				Protocol Type			
Hardware length		Protocol length		Operation Request:1, Reply:2			
Source hardware address							
Source protocol address							
Destination hardware address (Empty in request)							
Destination protocol address							

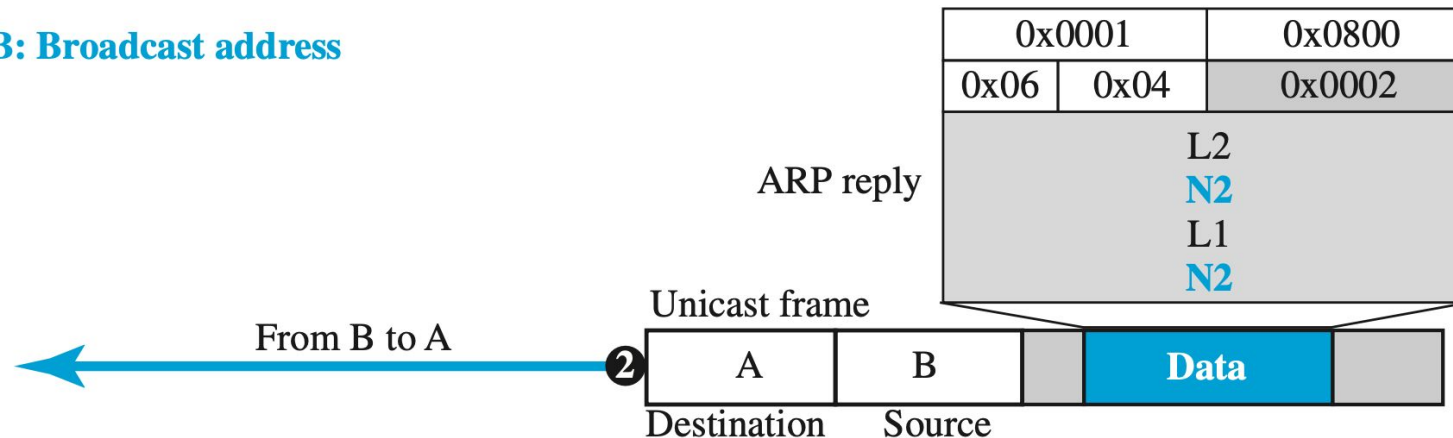
Hardware: LAN or WAN protocol

Protocol: Network-layer protocol

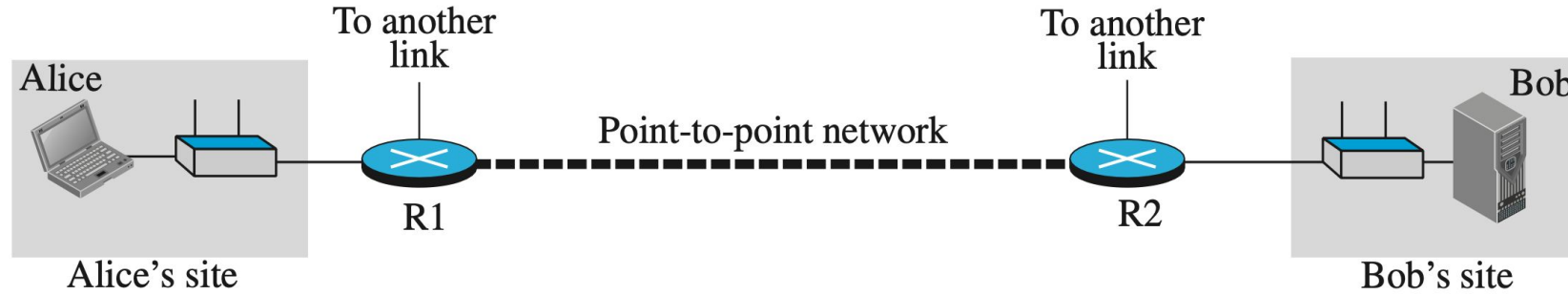
ARP Packet Example



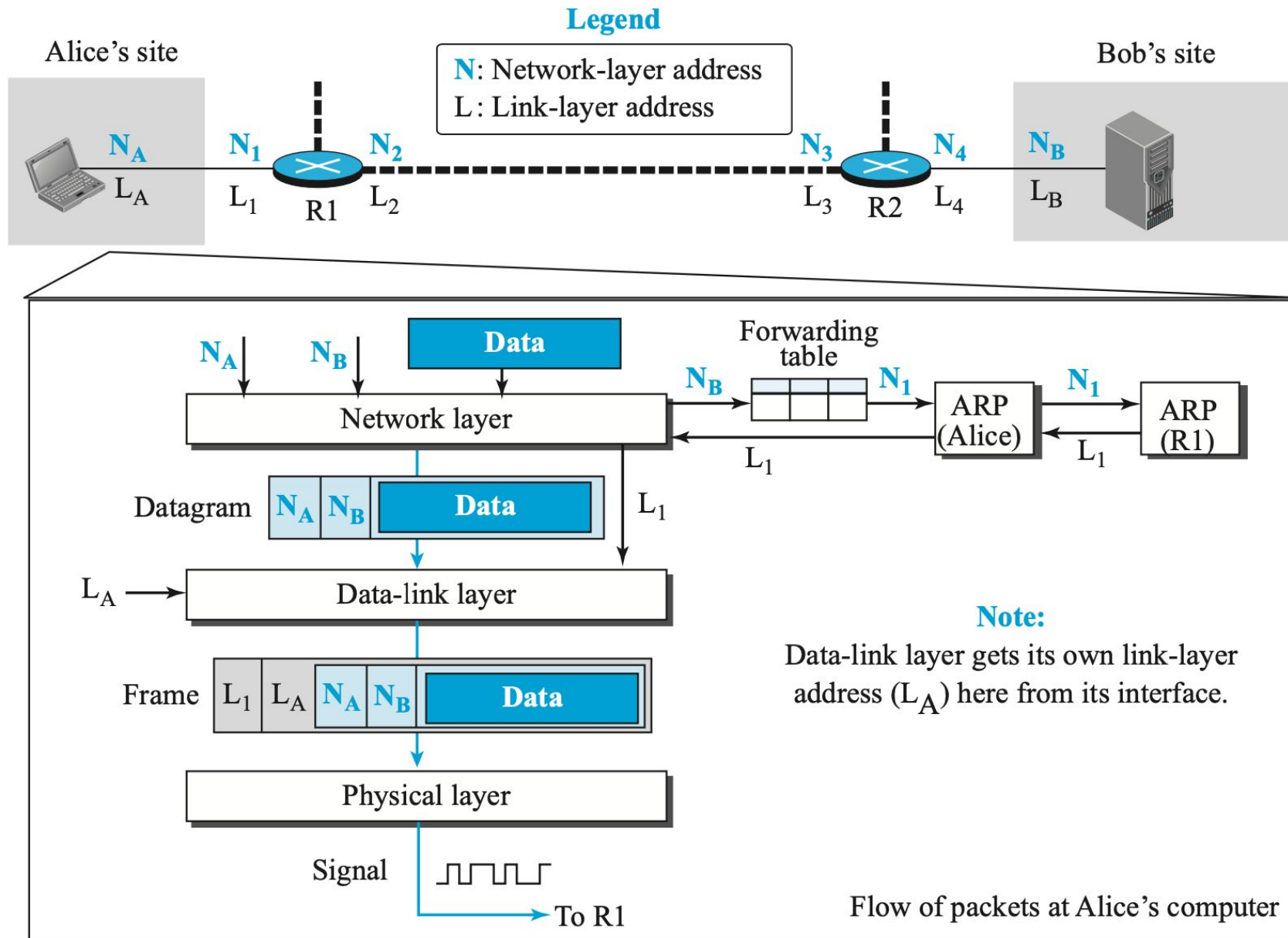
B: Broadcast address



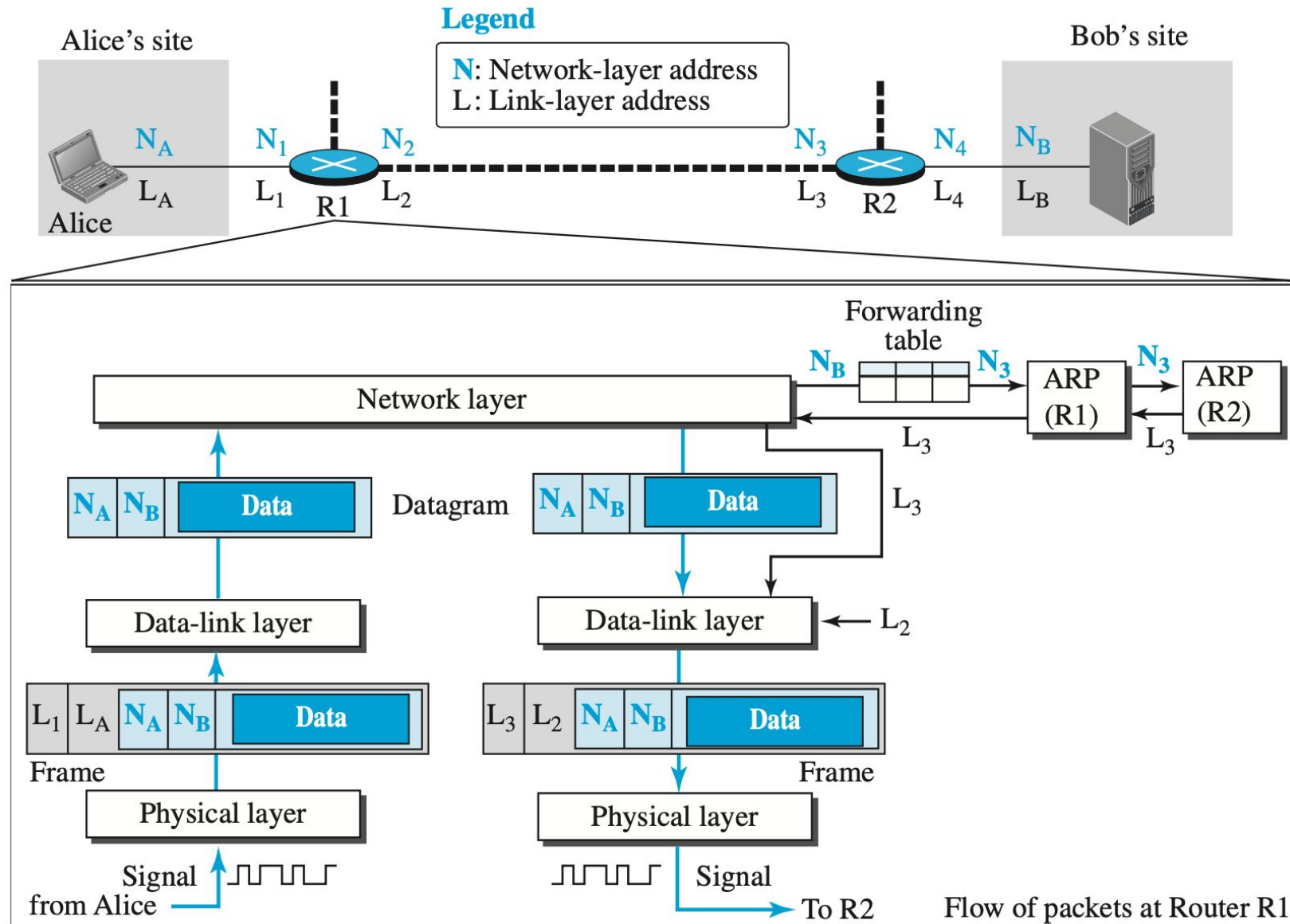
An Example of Communication



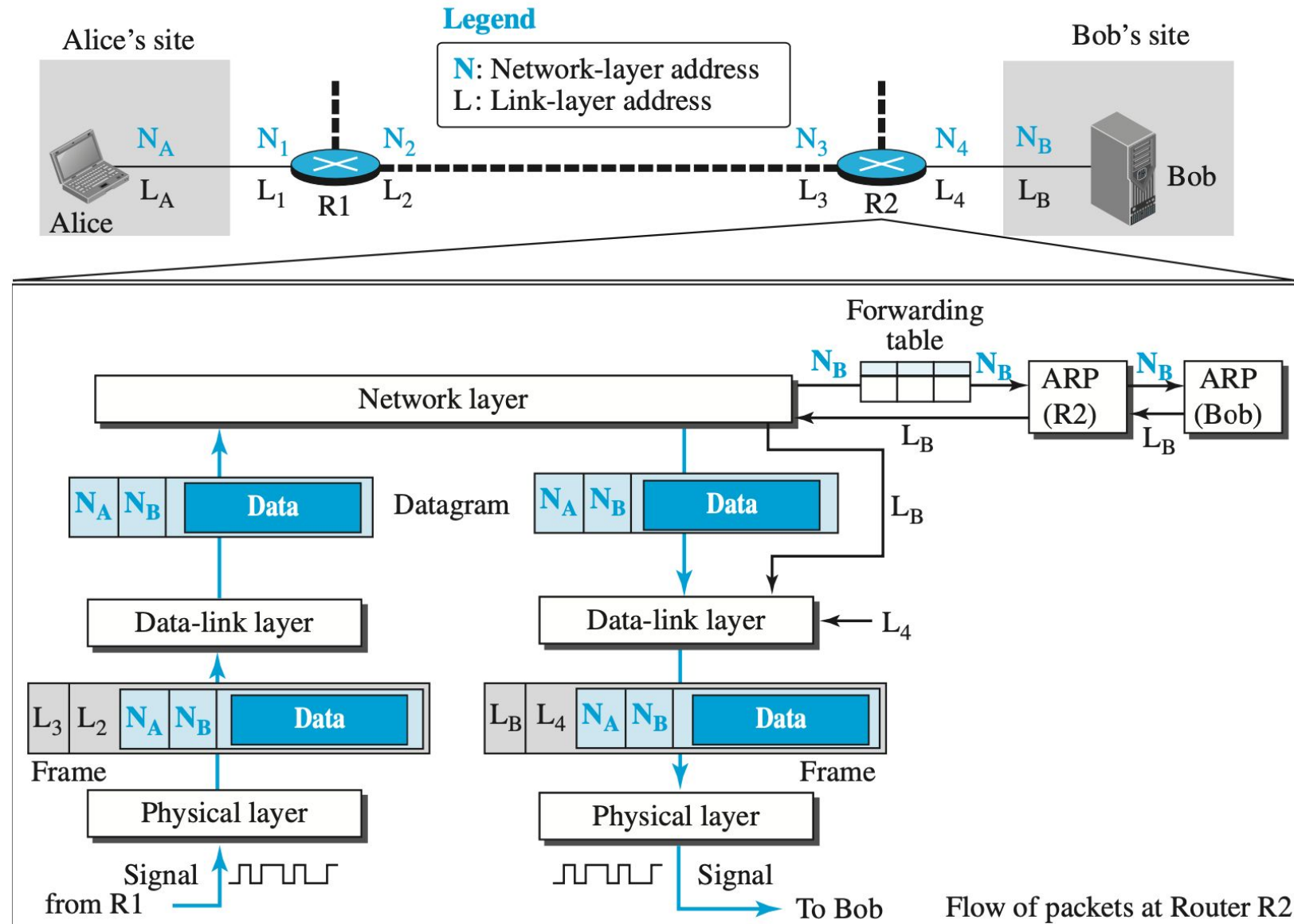
Activities at **Alice's** Site



Activities at Router R1



Activities at Router R2



Activities at Bob's Site

